

Research Article

How Consistent Are the Best-Known Readability Equations in Estimating the Readability of Design Standards?

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Abstract—Research problem: Readability equations are widely used to compute how well readers will be able to understand written materials. Those equations were usually developed for nontechnical materials, namely, textbooks for elementary, middle, and high schools. This study examines to what extent computerized readability predictions are consistent for highly technical material—selected Society of Automotive Engineers (SAE) and International Standards Organization (ISO) Recommended Practices and Standards relating to driver interfaces. **Literature review:** A review of original sources of readability equations revealed a lack of specific criteria in counting various punctuation and text elements, leading to inconsistent readability scores. Few studies on the reliability of readability equations have identified this problem, and even fewer have systematically investigated the extent of the problem and the reasons why it occurs. **Research questions:** (1) Do the most commonly used equations give identical readability scores? (2) How do the scores for each readability equation vary with readability tools? (3) If there are differences between readability tools, why do they occur? (4) How does the score vary with the length of passage examined? **Method:** Passages of varying lengths from 12 selected SAE and ISO Recommended Practices and Standards were examined using five readability equations (Flesch-Kincaid Grade Level, Gunning Fog Index, SMOG Index, Coleman-Liau Index, and Automated Readability Index) implemented five ways (four online readability tools and Microsoft Word 2013 for Windows). In addition, short test passages of text were used to understand how different readability tools counted text elements, such as words and sentences. **Results and conclusions:** The mean readability scores of the passages from those 12 SAE and ISO Recommended Practices and Standards ranged from the 10th grade reading level to about 15th. The mean grade reading levels computed across the websites were: Flesch-Kincaid 12.8, Gunning Fog 15.1, SMOG 12.6, Coleman-Liau 13.7, and Automated Readability Index 12.3. Readability score estimates became more consistent as the length of the passage examined increased, with no noteworthy improvements beyond 900 words. Among the five readability tools, scores typically differed by two grade levels, but the scores should have been the same. These differences were due to how compound and hyphenated words, slashes, numbers, abbreviations and acronyms, and URLs were counted, as well other punctuation and text elements. These differences occurred because the sources for these equations often did not specify how to score various punctuation and text elements. Of the tools examined, the authors recommend Microsoft Word 2013 for Windows if the Flesch-Kincaid Grade Level is required.

Index Terms—Ergonomics, human factors, plain language, readability, readability equations, standards, usability.

INTRODUCTION

George Klare defined readability in three ways, the last being the most recognized definition: “the ease of understanding or comprehension due to the style of writing” [1, p. 1]. Readability, as defined here, has to do with cognition and should not be confused with legibility—how easily one can

distinguish individual characters—which has to do with perception.

Many believe that readability can be estimated to a high degree of precision and accuracy by using equations based on linguistic statistics for the associated text. For example, Duffy noted that

There has been widespread application of readability formulas as a tool for defining plain English in the production of texts as well as in judging existing documents. [2, p. 113]

There is, however, evidence to the contrary described in detail later (e.g., [2] and [3]).

Since equations to compute readability are available on publicly accessible websites and incorporated into versions of Microsoft Word [4], the most widely used editing software, computer tools for computing readability are widely available. Belief in the utility of these equations, coupled with

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ready access to the tools to apply them, should encourage their use. Before doing so, however, one should examine the consistency of the analyses provided by these tools for the types of materials that engineers use. This precaution is important because engineering documents contain elements (headings, bulleted lists, equations, tables, figures, reference citations) that were often not included in the materials used to develop these equations [5]. This study examines the use of readability equations to estimate the readability of technical standards and, in particular, selected Society of Automotive Engineers (SAE) and International Organization for Standardization (ISO) Recommended Practices and Standards associated with driver interfaces.

Readability equations were originally developed for educational purposes. For example, to support the teaching of English in the US, state boards of education, local school boards, and superintendents selected textbooks for each grade in elementary, middle, and high schools. To make those decisions, they needed to know the grade reading level of textbooks being considered, levels computable using readability equations. Standardized reading tests (e.g., the McCall-Crabbs reading test) in the 1920s served as an initial basis for validating equations to predict the grade reading levels of textbooks. Thorndike's *The Teacher's Word Book* [6] was first published in 1921 and provided data on the frequency of use of the 10 000 most common words. This book is important because frequent words are more likely to be understood and, accordingly, word frequency is a factor in many readability equations, especially some of the earlier ones.

The first readability equation is associated with Lively and Pressey [7], with subsequent equations being developed in the 1930s, 1940s, 1950s, and 1960s. George Klare's landmark book *The Measurement of Readability* [1] was first published in 1963. By the 1980s, there were about 200 equations and more than 1000 studies published concerning readability equations. (For an overview of the history of this topic, see Dubay's "The Principles of Readability" [8] and "Unlocking Language: The Classic Readability Studies" [9].)

In the last few decades, probably beginning with the advent of the plain language movement, the emphasis has shifted from developing new equations primarily for textbooks to applying the concept of readability to a wider range of materials that are often more technical, such as patient information [10] and financial information [11].

Despite the popularity of readability equations, there are concerns about their limitations. Redish [12, pp. 47-48] listed eight concerns, including that equations "can only be used on prose passages," "do not give the writer any indication of how well organized the entire document is," and "do not measure typographic features—illustrations, type size, typeface, use of white space, attractiveness of presentation." Research on the readability of health-based literature [13] shows that (1) patients' reading skill; (2) aspects of the material, such as technical accuracy, format, learnability, and motivational messages; and (3) experiential or cultural factors of the target group should also be considered for effective communication, not just linguistic element statistics included in readability equations. In addition, different readability equations sometimes provide different scores for the same text [14]. Finally, most equations were developed decades ago, and they do not reflect changes in format (such as the increased use of headings and bullets), the role of text layout, or the inclusion of equations and code. This problem is particularly important for engineering texts, which contain tables and figures and have been given scant attention in the traditional readability literature. Kincaid's research [15] is a notable exception of focus.

Accordingly, the paper examines the following questions:

- RQ1.** Do the most commonly used equations give identical readability scores?
- RQ2.** How do the scores for each readability equation vary with readability tools?
- RQ3.** If there are differences between readability tools, why do they occur?
- RQ4.** How does the score vary with the length of passage examined?

LITERATURE REVIEW

The body of literature on the development and evaluation of readability equations is extensive. Given the specific purpose of this paper, this review focuses on a few of the most popular and accessible equations and on why the readability tools used to implement them can lead to inconsistent results.

The first section describes how literature was selected to investigate the topic of readability equations. Then, we present a detailed description of five common readability equations. Finally, we discuss prior research on computational inconsistency in readability, arguing that more

investigations into computational inconsistency are needed.

How Readability Equations Were Selected We selected five readability equations in common use [16] and for which multiple readability tools were readily available. Papers describing those equations were reviewed. Then, we searched commonly used electronic databases (such as Google Scholar, EBSCO, IEEE Xplore, ERIC) for research on readability by using the keywords “readability,” “readability equations,” “computational readability,” and “readability in technical communication.” Studies of particular interest were related to the development of readability equations, criticism of readability equations, and research on computational readability scores.

Description of Five Common Readability Equations This paper concerns five common readability equations [17], all of which can be calculated using multiple tools. The idea behind these equations is that the more complex or unusual (and numerous) the constituents of a text passage are, the more difficult the passage will be to read and understand.

Complexity can take the form of sentence-level factors, such as the number of words in a sentence, and word-level factors, such as the number of syllables in each word [17]. Furthermore, as was noted previously, frequently used words are more likely to be understood. However, syntax-related characteristics, such as the number of independent and dependent clauses, are not commonly considered.

Many readability equations have been developed using these measures [18]. The **Flesch-Kincaid Grade Reading Level** (equation 1) predicts a reading grade level of written materials [15, p. 14]. Kincaid et al. [15] developed the equation for the US Navy. The sample consisted of 531 US Navy-enlisted personnel from four training schools at two Navy bases. The scaled grade reading level of 18 test passages from Navy training manuals were calculated based on the criterion that 50% of subjects with a reading ability at a particular grade level had to score 35% or better on the cloze test. (In a cloze test, some words from a passage are removed and the participant replaces the missing words. The performance score is the percentage of correct replacements.) Test passages concerned topics such as temporary repairs to the hull and sea-wave observations. Using these data and multiple regression analysis, the parameters for the

Flesch grade reading level equation were determined. No correlation coefficient or R^2 value was presented in the paper to quantify how well the equation predicted the data:

$$0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59. \quad (1)$$

In the Kincaid analysis, contractions (e.g., don’t), digit strings (e.g., 1,000,000), and hyphenated words (e.g., second-grade) are counted as one word. For sentences, a grammatically independent sentence or clause is considered as one sentence. Finally, for syllables, digits were treated as if they were spelled out (“1918” is four syllables) (see [15, pp. 39-40] for the rules for counting words, syllables, and sentences). Kincaid et al. [15] did not specify sample size requirements, but the passages of material selected as representative had a mean length of 170 words.

The **Gunning Fog** Index (equation 2) is similar to the Flesch equation in that it also considers syllables and sentence lengths [19, pp. 38-39]. A Fog score of 5 is readable, 10 is hard, 15 is difficult, and 20 is very difficult. Gunning found that readability could be predicted using only two statistics: the mean sentence length and a hard-word factor. According to Gunning, the fraction of words of three syllables or more was the best indicator of the hard-word factor. To provide perspective, he found that only one out of 25 in Dale’s list of 3000 most familiar words had more than two syllables. In contrast, of those words not among the 20,000 most often used, two out of every three were three syllables or more [19]. Gunning’s original descriptions of his equation did not appear in a scientific journal, so fewer details about how the equation was developed are available [20].

$$0.4 \left[\left(\frac{\text{words}}{\text{sentences}} \right) + 100 \left(\frac{\text{complex words}}{\text{words}} \right) \right]. \quad (2)$$

In the Gunning Fog Index, “Foggy” (complex) primary words are those that contain three or more syllables. However, proper names, combinations of short words or verb forms that become three syllables by adding -ed or -es are not counted as complex words [19]. Although rules for counting sentences are not explicitly given, Gunning provided examples. Interestingly, Gunning counted a compound sentence with two independent clauses as two sentences, with opinions about the merits of counting them in that manner differing.

The **SMOG Index** (equation 3) provides a US school grade level [21, p. 643]. The SMOG Index was developed as a quick and easy equation to estimate readability based on two statistics: the number of sentences of the selected article and the number of words with three or more syllables. McLaughlin [21] developed the equation based on a regression analysis of 390 passages included in the 1961 edition of the *McCall-Crabbs Standard Test Lessons in Reading*, resulting in a correlation of 0.99 between the predicted and actual test scores given in the book. Note that the McCall-Crabbs Test was first introduced in 1925, and it was revised in 1950 and again in 1961. However, subsequent research found that the 1961 version was out of date and needed to be redone, a fact that suggests that formulas based on this test may no longer be accurate [22].

To compute the SMOG Index, McLaughlin [21] suggested counting 30 sentences (10 consecutive sentences near the beginning, 10 in the middle, and 10 near the end). The only specific guidance for counting words was to count three- or more-syllable words by the way they are normally pronounced. Furthermore, to simplify the equation, McLaughlin [21] recommend replacing 1.0430 in the Index with 1 and 3.1291 with 3 for simplicity:

$$1.0430 \sqrt{\frac{\text{number of polysyllables}}{\text{number of sentences}} \times 30} + 3.1291. \quad (3)$$

Coleman-Liau Index (equation 4), a less commonly used equation, also provides a US school grade level (derived from [23, p. 284]). The Coleman-Liau Index was developed to facilitate scoring using mechanical counting devices, an important innovation. There was no specific sample size requirement for their equation. Coleman and Liau [23] used 36 150-word passages with cloze percentages scaled by Miller and Coleman [24] as samples to develop predicted cloze percentages. Then, they developed a second regression equation predicting the grade levels from the cloze percentages which, combined with the prior regression analysis, led to

$$0.0588L - 0.296S - 15.8. \quad (4)$$

L is the average number of letters per 100 words and S is the average number of sentences per 100 words.

Counting letters is straightforward. However, as is described elsewhere, what constitutes a sentence for the purpose of counting may be ambiguous.

The **Automated Readability Index** (equation 5) also provides a grade level and was developed to assess the readability of written materials used in the US Air Force [25, p. 8]. The goal was to add a sensor to a typewriter so that the grade level could be computed automatically as material was typed. That material can be more technical than grade school textbooks. A 10-page sample was suggested for computing the Automated Readability Index [25, p. 12]. Senter and Smith developed the Automated Readability Index using samples of text from 24 books with assigned reading grade levels. For each sample, the mean number of characters/word and words/sentence was determined. Using regression analysis, an equation was developed relating those two factors to the reading grade level. The reported R^2 was 0.98, quite high:

$$4.71 \left(\frac{\text{characters}}{\text{words}} \right) + 0.5 \left(\frac{\text{words}}{\text{sentences}} \right) - 21.43. \quad (5)$$

At a surface level, determining readability seems straightforward. Count the number of words, syllables, or other sentence characteristics, add and multiply them together as indicated, and one is done. However, besides the inherent limitations of readability equations described in the Introduction, real passages have hyphenated words, dashes, and other elements that make determining what is a word (and counting syllables in a word) in a readability equation subject to inconsistent interpretation.

Prior Research on Computational Inconsistency

Readability scores are invariably now computed using software, not calculated by hand. The common impression, one that the authors held before this study, was that *computers give highly consistent predictions of the readability of a passage of text*, so given a sufficient sample, there should be no differences between readability tools computing the readability score using the same equation and very small differences between equations. After all, what constitutes a syllable, word, and sentence, the units of analysis for readability equations, is precisely defined, and computers do not make mistakes. There is, however, some not-well-known evidence to the contrary. Mailloux et al. [26] examined implementations of the Flesch Kincaid Adjusted Reading Level, Flesch Reading Ease, and Gunning Fog readability equations on four websites based on 28 pieces of nationally distributed HIV/AIDS educational literature. Among the four sites, the range of mean Gunning Fog scores was about 3.0, the range of mean Flesch-Kincaid scores was about 1.7, and the range of Reading Ease

scores was about 0.6. Mailloux et al. did not indicate why the differences occurred.

In an obscure reference discovered after Mailloux et al.'s research was completed, Sydes and Hartley [27] conducted five experiments to examine the readability predictions from four applications: Grammatik4, Read, Word for Windows 2, and Word for Windows 6. Although these applications are now obsolete, the findings concerning them are not. First, Sydes and Hartley examined passages from articles in the *British Journal of Psychiatry* and found that mean Flesch Reading Ease scores differed by up to 10 between applications. Subsequently, they examined the causes of differences, including punctuation (brackets, square brackets, quotation marks, colons, semicolons, slashes, and hyphens), with the effect of colons and semicolons depending on the application, but not other punctuation. All of the applications treated hyphenated words as one word, though there were differences in syllable counts. There were also differences in how digits (e.g., 11 versus 1111), abbreviations (e.g., D.N.A. versus DNA), and dates were treated as a function of format (e.g., 1st January 1995 versus 1/1/1995). Finally, there were differences in terms of how rare and nonsense words were treated. Given these issues, Sydes and Hartley suggested that word-based equations (such as the Gunning Fog Index) could be more appropriate for more difficult text than syllable-based equations (such as the Flesch Reading Ease equation).

METHODOLOGY

This section provides readers with the methodology used in this paper. First, we state our research questions related to readability equations and computational scores. Second, we discuss our choice of research methodology. Then, we present the methods we used to collect data in terms of tools and samples for each of four parts: (1) a readability analysis of selected ISO and SAE standards, (2) an analysis of the differences between five readability tools, (3) an analysis of different counting mechanisms used in the five readability tools, and (4) an analysis of the impact of sample passage length. Finally, we discuss the data-analysis process.

Research Questions Based on previous research, we hypothesize that different computational readability tools and different readability equations can lead to different readability predictions. However, except for Sydes and Hartley [27], an

obscure study nearly two decades old, most of the prior studies of prediction inconsistency did not systematically investigate why differences occurred.

Furthermore, in the years since many of these equations were developed, engineering texts have evolved, making greater use of headings and bulleted lists, and software has become important, so code frequently appears in engineering materials. Accordingly, our paper examines the following questions:

RQ1. Do the most commonly used equations give identical readability scores?

RQ2. How do the scores for each readability equation vary with readability tools?

RQ3. If there are differences between readability tools, why do they occur?

RQ4. How does the score vary with the length of passage examined?

Choice of Research Methodology We chose a quantitative over a qualitative approach primarily because we could manipulate and analyze the readability scores (numeric values) given by different readability analysis tools. The statistical tools available to process a large set of data were rich and credible. We believed that this quantitative analysis could give us insight into understanding the questions listed before.

How the Data Were Collected: Using those online readability tools, readability scores were calculated for the material uploaded. A Python web crawler automatically collected and saved those scores in a Microsoft Excel file. Sample materials tested for readability scores included ISO and SAE standards [28]–[39], short segments of text (test cases) and common text fragments, test passages from Kincaid et al.'s *Derivation of New Readability Formulas for Navy Enlisted Personnel* [15], and text passages from Gunning's *The Technique of Clear Writing* [19].

Methodology in Readability Analysis of Selected ISO and SAE Standards: As part of an industry-funded project to summarize and assess SAE and ISO Recommended Practices and Standards relating to driver interfaces for warning systems [40], 12 of them (Table I) were examined in detail. In the follow-on activity described in this paper, readability scores were calculated for those 12 highly technical documents.

To prepare the 12 documents for analysis, all titles, figures, tables, equations, headings, and references were removed, as those elements were not intended

TABLE I
SAE AND ISO STANDARDS AND RECOMMENDED PRACTICES EXAMINED

	Document no.	Partial Title	No. of Pages
SAE	Recommended Practice J2395 [28]	ITS In-Vehicle Message Priority	7
	Recommended Practice J2396 [29]	Definitions & Measures for Visual Behavior	13
	Standard J2399 [30]	ACC Characteristics & User Interface	11
	Information Report J2400 [31]	FCW Human Factors: Operating Characteristics & User Interface Requirements	23
	Recommended Practice J2802 [32]	Blind Spot Monitoring System: Operating Characteristics & User Interface	10
	Information Report J2808 [33]	Road/LDW Systems: Human Interface	10
	Information Report J2830 [34]	Process for Testing Icon Comprehension	14
	Information Report J2831 [35]	Recommendations for In-Vehicle Alphanumeric Text Messages	41
ISO	Standard 15005 [36]	Dialogue Management Principles & Compliance Procedures	22
	CD Standard 15007-1 [37]	Measurement of Driver Visual Behavior - Part 1: Definitions and Parameters	17
	Standard 15008 [38]	Specifications & Tests for In-vehicle Visual Information	26
	Standard 17287 [39]	Procedure: Assess Suitability for Use While Driving	36

to be considered by readability equations. If anything, not considering them is a weakness of readability equations because those elements are important for readers to understand each document.

Following good scientific practice of double-checking results for accuracy, readability scores were obtained from four websites—readability-score.com, read-able.com, online-utility.org, and editcentral.com. In searching for “readability calculator” using Google, these four were among the first listed that provided readability scores for the five most common equations: Flesch-Kincaid Grade Level, Gunning Fog Index, SMOG Index, Coleman-Liau Index, and Automated Readability Index. Microsoft Word 2013 for Windows was used only to test Flesch-Kincaid scores, as it did not provide results for the other equations. Note that editcentral.com used approximations for Coleman-Liau Index parameters (5.89 instead of 5.88, 30.0 instead of 29.6; see [41]). However, those approximations should have only a minimal effect on the final readability score.

Methodology for the Analysis of the Differences Between Five Readability Tools: The test material consisted of 18 passages from Kincaid et al.’s *Derivation of New Readability Formulas for Navy Enlisted Personnel* [15] and 22 passages from Gunning’s *The Technique of Clear Writing* [19].

- (1) The 18 Kincaid passages had 3067 words in total, with a mean length of about 170 words. As noted previously, these passages were originally selected from US Navy training manuals.
- (2) The 22 Gunning passages had 5407 words in total, with a mean length of about 250 words.

Gunning Fog Indexes were determined for the passages from Gunning.

In addition to using the four websites to determine readability scores for the five common equations, Flesch-Kincaid Grade Levels were determined using Microsoft Word 2013 for Windows.

Papers describing the SMOG index, Coleman-Liau Index, and Automated Readability Index did not provide test samples, so those equations were not examined.

Methodology for the Analysis of the Different Counting Mechanisms Used in Five Readability Tools Because readability equations usually use linguistic element statistics (often counts) to calculate grade levels, how each site identified those elements—sentences, words, syllables, complex, and polysyllabic words—was examined. The same four online tools, together with Microsoft Word 2013 for Windows, were used to compute linguistic statistics of several specific test cases. Some of the websites did not provide all of the statistics used in the grade reading-level calculation, such as the number of words with more than three syllables. Fortunately, those statistics could still be derived from the readability scores provided by the websites. To develop initial hypotheses, readability scores sections of these documents were computed for each equation × website combination, and the scores were compared. Where differences in the scores and measures that contributed to the scores appeared, subsets of the texts were then used to narrow down the sources of the differences. Specifically, very short passages of text (test cases) were examined with only a single anomaly (e.g., numbers as

TABLE II
EXAMPLES AND TEST CASES EXAMINED

Statistic	Examples	Test Cases
Number of sentences	Count a compound sentence by counting the number of independent clauses Correctly ignore quotation marks when counting sentences Ignore periods in phrases instead of regarding them as the ends of sentences	"... in both hemispheres, but the 'explosive warming' prior to ...", "TICSs are intended for use in a moving vehicle; TICSs ..." "Good afternoon. Commander," "Good evening. Chaplain." "e.g.", "Mr.", "3.24"
Number of words	Count possessive as one word Count a hyphenated word as one word Count two words separated by a slash as two words Ignore en and em dashes during word counting Count syllables of symbols and figures as they are normally pronounced Count a URL as one word	"subject's", "driver's" "in-vehicle", "copy-edit" "ISO/SAE", "and/or" "33-47", "June 30-July 2", "No one—not even the president of the company—..." "1918", "R.F.D." "www.apple.com", "http://www.google.com"
Number of syllables or complex words	Count proper nouns as simple words, even with more than three syllables Count combinations of easy words or hyphenated words as simple words Count two-syllable verbs made into three with -es and -ed endings as simple words	"Microsoft", "Chicago" "butterfly", "copy-edit" "Information is gathered ..."
Number of characters	Count punctuation marks as characters Ignore dots used in numbering a list	"driver's", "in-vehicle", "display/TICS" "1. Scope—This SAE Recommended ..."

digits). Table II shows measures of interest and test cases used for each measure.

Some test cases were excerpted from documents examined in this report. Others were common text fragments used in everyday life.

Methodology for Analysis of Impact of Sample Passage Length

The same 12 ISO and SAE documents (see Table I) were used to analyze the impact of sample passage length in determining readability scores, and random passages with 100 words, 200 words, 500 words, and 900 words, as well as the entire document (minus unscorable sections), were selected from those documents as test cases. The same four online tools (readability-score.com, read-able.com, online-utility.org, and editcentral.com) were used to compute the mean readability scores for the different equations, and standard deviation among different online tools was computed for each passage length.

To present the data collected, we used descriptive statistics, such as the mean, range, and standard deviation. Since we had two experimental conditions for the first two tests, we used a two-way ANOVA to test the significance of different readability scores given by different formulas or

different readability tools. In the test of sample size, the interquartile range of the standard deviation was used to represent the consistency of readability scores given by online readability tools.

RESULTS

The purpose of this section is to answer our research questions using quantitative analyses of the readability scores, and to discuss what we found.

RQ1. Do the most commonly used equations give identical readability scores?

As shown in Fig. 1, for each of the 12 documents examined, there are practical differences among the five readability equations examined. In a two-way ANOVA (document $\times$ equation) of the grade-level scores, there were statistically significant differences between equations ($F(4, 44) = 63.77$, $p < 0.001$). Note that these analyses were not based on samples of text, but on all of the text that was available to be analyzed. We assumed that tenths of a grade are meaningful. The mean grade reading levels computed across sites were Flesch-Kincaid Grade Reading Level 12.8, Automated Readability Index 12.3, SMOG Index 12.6, Coleman-Liau Index 13.7, and Gunning Fog Index 15.1.

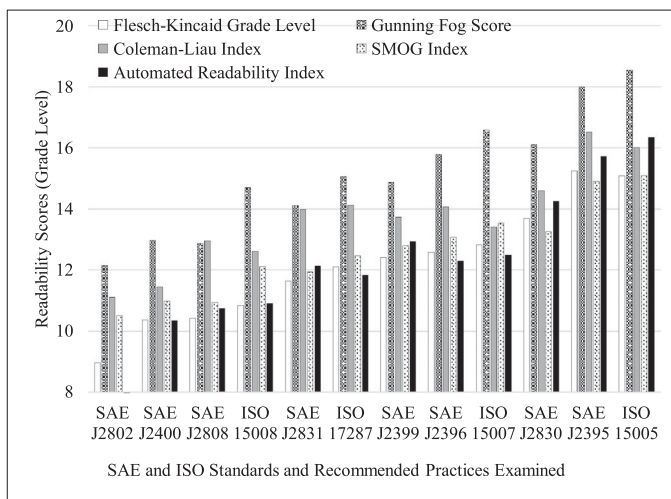


Fig. 1. Mean readability scores (averaged across five readability tools) from five equations.

The differences are not surprising given previous research [14]. The Gunning Fog Index led to the highest estimated grade reading level for every document except one (SAE J2808). There were many instances where the Gunning Fog Score was three grades greater than that estimate by other equations. Thus, readability scores of the same piece of text would vary depending upon the equation selected. Furthermore, the decision of whether a text was acceptable for a particular audience or not would depend upon the equation chosen.

In addition, the level that is acceptable deserves some thought. According to these calculations, SAE J2802, on average, is somewhat less than a 10th grade reading level, and ISO 15005 is about 15th, or third year of college. If these documents are intended primarily for college graduates (often engineers), one could consider these grade-reading levels to be adequate for the intended audience. This assumption is consistent with Gazni's conclusion [42] that academics tolerate apparently difficult text. However, one must keep in mind that these documents are used internationally, and many users are not native English readers. What may be appropriate for native English-speaking engineers that are college graduates may not be appropriate for the global engineering community.

RQ2. How do the scores for each readability equation vary with readability tools?

Though not as large as the differences between equations, averaging the data across equations and examining them using readability tools indicates there are practical differences between readability tools (Fig. 2). These mean grade reading levels for

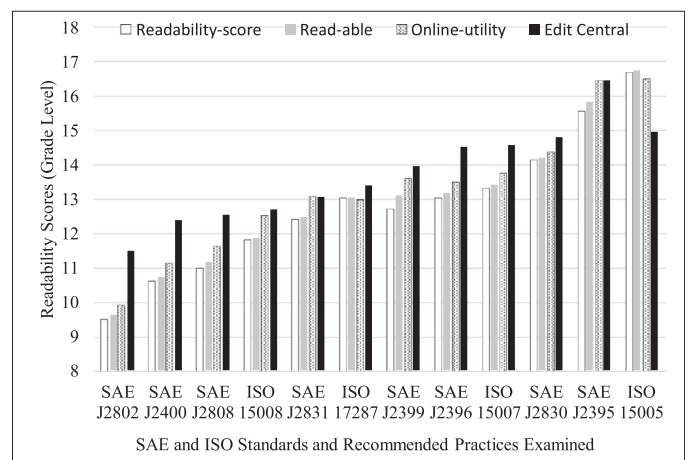


Fig. 2. Mean readability scores (averaged across five equations) from four online readability tools.

the same text differed by two grade levels, a substantial difference. The mean grade reading levels were readability-score.com 12.8, read-able.com 12.9, online-utility.org 12.3, and editcentral.com 13.7. In fact, a two-way ANOVA (document $\times$ readability tools) of the grade-level scores found statistically significant differences between readability tools ($F(3, 33) = 9.14$, $p < 0.001$). The editcentral.com scores were consistently greater. Thus, not only does the equation matter in determining whether a document has an acceptable computed readability level, but so too does the readability tool used.

If one were to assume that calculating grade reading levels using these equations is a well-defined procedure and that computers do not make mistakes, then these findings do not make sense. This finding of between-site differences led to a series of evaluations to explain those differences.

How Well Do Grade Reading Levels Calculated By the Readability Tools Agree With the Original Levels Given By the Developers of the Equations?

Fig. 3 shows the Flesch-Kincaid Grade Level computed by the four websites and Microsoft Word for the 18 test passages in *Derivation of New Readability Formulas for Navy Enlisted Personnel* [15]. Note that editcentral.com scores were consistently lower than those produced by the other readability tools, and those scores had the lowest correlation (0.91) with the grade levels given in Kincaid et al.'s paper. Other readability tools, particularly Microsoft Word, provided grade reading levels that were closer to the original published values. This result could be due primarily to how editcentral.com counts (actually miscounts) syllables.

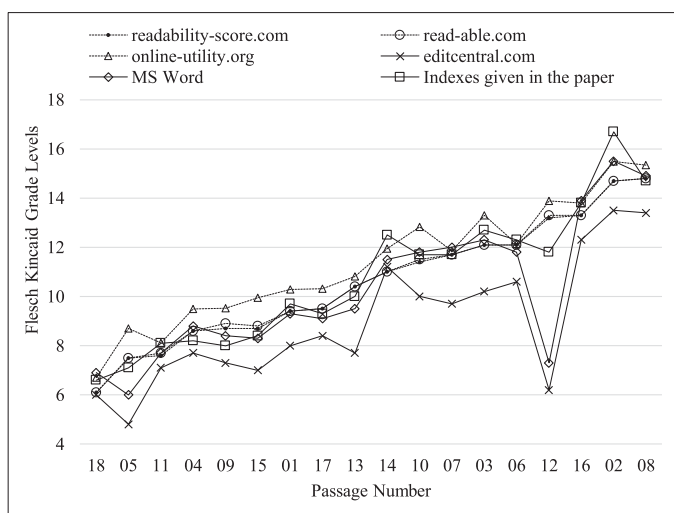


Fig. 3. Flesch-Kincaid Grade Levels for test passages used in Kincaid et al. [15].

Nonetheless, the data are consistent, and the overall correlations between the original scores and each of the tools are 0.96 for readability-score.com, read-able.com, and online-utility.org, 0.92 for Microsoft Word, and 0.91 for editcentral.com. Note that Microsoft Word makes the fewest counting mistakes (as shown in Table III). Part of the problem is that the original scores Kincaid gave were based on criteria in Table IV, some of which are vague. Therefore, discrepancies are expected.

For the 12th passage, there was a huge grade difference (7.7) between scores given by editcentral.com and the original Flesch-Kincaid score. An example follows:

For example, there are fiber manropes on gangways; foot ropes on hammocks (and formerly on the yards of sailing ships); bolt ropes on sails and other canvas; ridge ropes on awnings (but they are usually of wire); dip ropes for passing some object under, outside of, or around another; and bull ropes for heavy heaving without benefit of a purchase (tackle).

There are many semicolons in the paragraph. The grade level reported in this paper suggests that semicolons were not considered as the end of a sentence, whereas editcentral.com regarded semicolons separating two independent clauses and counted them as two sentences.

In contrast, Gunning Fog scores (Fig. 4) were more varied between websites than those for Flesch. Specifically, the correlations between the original passage scores (from Gunning) and those provided

by readability tools were 0.87 for readability-score.com and readable.com, 0.85 for edit-central.com, and 0.79 for online-utility.org. For the Gunning Fog Index, no readability tool appears to provide scores that are vastly different from others. The Gunning Fog Index counts compound sentences as multiple sentences, but the counting mechanism of the online tools appears to be different. As compound sentences are quite common in English, this fact is a problem.

RQ3. If there are differences between readability tools, why do they occur?

In the section above describing the five readability equations examined, how each equation counts different linguistic elements is briefly mentioned. Based on the information in the literature describing those equations, Table IV gives a detailed summary of how equation developers count different text segments shown in Table II. Many equations simply give a mathematical expression with vague criteria for how to count those factors used in the expression. Sometimes they even contradict each other. For example, Gunning [19] counted a compound sentence with two independent clauses as two sentences, but other equations would count a compound sentence as only one sentence.

Table III shows how different readability tools count different text segments presented in Table II. By comparing Tables III and IV, one can see how the criteria used by each readability tool to determine what is a syllable, word, and sentence influence the readability scores for each equation.

Generally, all of those tools had problems in counting sentences, words, and characters as they missed many special cases. In examining those counts, keep in mind that there were differences between equations in terms of how linguistic elements were determined (counted). Nonetheless, among the tools, Microsoft Word 2013 for Windows scores tended to be the most consistent with published counting criteria. In contrast, editcentral.com had many problems with counting syllables, leading to erroneous estimates of grade reading levels.

RQ4. How does the score vary with the length of passage examined?

As shown in Figs. 3 and 4, differences in grade reading levels for passages were typically two grade levels. However, the approach used initially was to examine the entire document minus text elements (e.g., tables) for which readability equations were not intended. How much does the sample size matter?

TABLE III
EQUATION DEVELOPERS' CRITERIA FOR COUNTING TEXT SEGMENTS

Statistic, Example	Equation				
	Flesch-Kincaid	Gunning Fog	Coleman-Liau	SMOG	Automated Readability
Number of Sentences					
Counts a compound sentence by counting the number of independent clauses	×	√	?	?	?
Ignores quotation marks	√	√	√	√	√
Ignores periods in a phrase not leading to the end of a sentence	√	√	√	√	√
Number of Words					
Counts possessive as one word	√	?	?		√
Counts a hyphenated word as one word	√	√	?	√	√
Counts two words separated by a slash as two words	×	?	?	?	×
Ignores en and em dashes	×	?	?	√	×
Counts syllables of symbols and figures by the way they are normally pronounced	√	×		√	
Counts URL as one word	?	?	?	?	?
Number of Syllables or Complex Words					
Counts proper nouns with more than three syllables as simple words		√		×	
Counts combinations of easy words or hyphenated words as simple words		√		×	
Counts two-syllable verbs made into three with -es and -ed endings as simple words		√		×	
Number of Characters					
Counts punctuation marks as characters			?		√
Ignores dots used in numbering a list	?	?	?	?	?

Note: √ = follows the rule; × = doesn't follow the rule; ? = unknown usage (not given by equation developers); blank = that factor is not needed in the equation (for example, Flesch-Kincaid does not count number of simple or complex words).

To address this question, a new test was conducted. Fig. 5 shows standard deviations among readability tools for each passage length. Since there were 12 different documents, for each sample size there were 12 standard deviations. Thus, the dispersion of the standard deviations represents the consistency of different readability tools for each sample size.

Notice that the interquartile range of the standard deviations appears to be about the same for passages of 100 and 200 words, less for 500 words,

and even less for 900 words or more. Keep in mind that the Flesch-Kincaid Grade Reading Level passages [15] were based on samples of 170 words, McLaughlin [21] suggested 30 sentence samples, and 10-page samples were suggested for the Automated Readability Index. If one assumes a sentence length of 20 words, probably reasonable for midrange (high school) readability, and 300 words on a typed page, then the three sample recommendations from the literature are 170, 600 (= 30 × 20), and 3000 (= 300 × 10) words, a huge range. It could be that the recommended word

TABLE IV
HOW DIFFERENT READABILITY TOOLS USE DIFFERENT COUNTING CRITERIA

Statistic, Example	Readability tools				
	readability- score.com	readable. com	online- utility.org	editcentral.com	Microsoft Word
Number of Sentences					
Counts a compound sentence by counting the number of independent clauses				% (only regard semicolon as the end of the sentence)	
Ignores quotation marks	√		√	√	√
Ignores periods in a phrase as the ends of a sentence				% (only recognize "e.g" but not "e.g.")	√
Number of Words					
Counts possessive as one word		√	√	√	√
Counts a hyphenated word as one word				√	√
Counts two words separated by a slash as two words	√				√
Ignores en and em dashes	√				√
Counts syllables of symbols and figures by the way they are normally pronounced					√
Counts a URL as one word					% (counts "http://www.google.com" as two words)
Number of Syllables or Complex Words					
Counts proper nouns with more than three syllables as simple words	% (count a capitalized word as a simple word even if it is not a proper noun)	% (count a capitalized word as a simple word even if it is not a proper noun)			*
Counts combinations of easy words or hyphenated words as simple words	% (Count a hyphenated word as two words)	% (Count a hyphenated word as two words)	% (Count a hyphenated word as two words)		*
Counts two-syllable verbs made into three with -es and -ed endings as simple words	√	√	√	√	*
Number of Characters					
Counts punctuation marks as characters			√	√	√
Ignores dots used in numbering a list					√

Note: √ = implemented, % = partially implemented, blank = not implemented.

* Microsoft Word provides only Flesch-Kincaid grading level, which doesn't count numbers of simple or complex words.

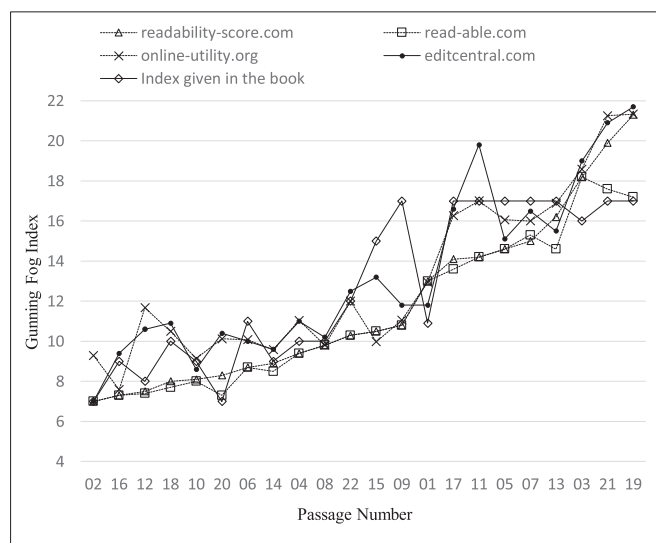


Fig. 4. Gunning Fog Indexes for test passages used in The Technique of Clear Writing [19].

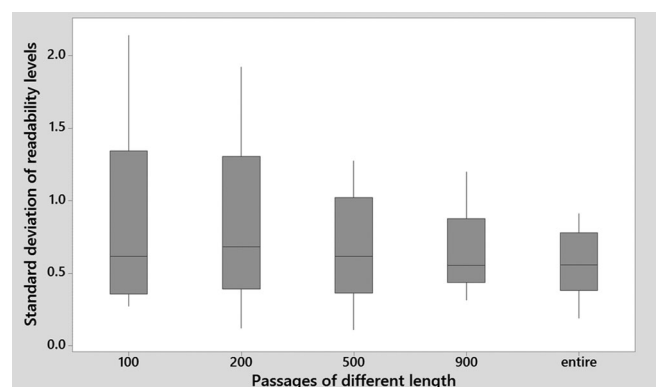


Fig. 5. Boxplot of standard deviations among different online tools ($N = 12$ for each category).

samples size varies with the equation, but given that they all include similar measures, that explanation does not seem likely.

What these data suggest is that somewhere between 500 and 900 words, the readability estimates stabilize. When analyzing samples with fewer words, a single unusual text structure (such as in the 12th passage) is more likely to lead to an unrepresentative score. Since the effort to examine an entire document, at least for standards and recommended practices, is not substantially greater than that to examine a long passage, computing readability scores using entire documents is recommended. However, even with this approach, scores between websites can vary substantially.

CONCLUSIONS, LIMITATIONS, AND SUGGESTIONS FOR FUTURE RESEARCH

Conclusions We offer five conclusions based on the findings of our study.

Do the Most Commonly Used Equations Give Identical Readability Scores?: No. Different equations compute different readability scores for the same text passages. The range of the mean grade reading levels among ISO and SAE Recommended Practices and Standards computed across sites was 2.3 grade levels. The differences are often even larger for an individual document.

Do Online Readability Tools That Calculate Readability Equations Give the Same Estimates, and If There Are Differences, How Large Are They?: Our results indicate considerable differences in the grade levels calculated by different common computerized readability tools calculating readability equations (Flesch-Kincaid Grade Level, Gunning Fog Score, Coleman-Liau Index, SMOG Index, Automated Readability Index). Using 12 ISO and SAE Recommended Practices and Standards as test materials, the grade reading levels (averaged across five equations) for the same text differed by as much as 2.0 grade levels, depending upon the online readability software used. The differences are even larger among different equations, sometimes as much as four grade levels.

Why Do Readability Scores Using the Same Equation Differ Between Readability Tools?: It would seem that defining a syllable, word, or sentence is straightforward, and every tool that calculates these readability equations should give the same score for the same text passage. That result did not occur, and the differences were substantial. Differences in sentence counts resulted from the way colons, semicolons, and other punctuation were treated. Differences in word and syllable counts resulted from the way hyphens, contractions, digits, dates, acronyms, abbreviations, and other text elements were treated. This problem occurs because the way that the number of words, syllables, and sentences is to be determined is sometimes vague and the rules used to count them vary between equations. There are no universal counting rules, at least for these equations.

How Does the Score Vary With the Length of Passage Examined?: For ISO and SAE passages of 200 words or greater, increasing the sample size decreased the interquartile range of the standard

deviation of estimates between sites, with the estimates continuing to decrease up to about 900 words. Therefore, samples used for readability estimates of technical documents, such as standards, should be somewhere between 500 and 900 words, with 900 words being the safe estimate. However, the effort required to examine the entire document is usually not much greater than that required for 900-word samples. Accordingly, we recommend using the entire document to reduce the risk of incorrectly estimating the grade reading level.

Should These Equations Be Used to Assess the Readability of Technical Material?: We would argue yes because the equations do provide useful insight, but equation scores should not be treated as precise estimates—for example, considering differences in tenths of a grade as being meaningful. Just because a computer produces a value does not mean that the value is accurate because accuracy depends on the rules the computer program follows. As was shown here, differences in interpretation of what constitutes a syllable, word, and sentence led to differences in readability scores typically ranging over two grade levels. Finally, keep in mind that what these readability equations do not consider can be a significant portion of an engineering document—headings, bulleted lists, equations, code, figures, and tables.

Recommendation The authors do not have a strong preference for using any particular readability equation. In fact, all readability equations used in this paper lack clear criteria for counting punctuation and text elements.

However, we would suggest using Microsoft Word 2013 for Windows if the Flesch-Kincaid Grade Level is needed because Microsoft Word 2013 makes the fewest errors in counting linguistic elements. Unfortunately, Word 2013 does not provide other readability scores, so calculating the mean score using multiple equations to lessen the effects of erroneous assumptions of particular equations is not possible in Microsoft Word, so other applications must be used.

Of course, the larger issue is whether these SAE and ISO standards are too difficult to read. The primary audience for these documents is engineers, and typically they have four years of college, a grade-16 education. However, as automotive engineering becomes more global, an increasing fraction of automotive engineers may not use English as their native language. Furthermore, technicians who do not have a college education also use these standards. Thus, until information becomes available on the grade reading level of these individuals, this larger question cannot be answered.

Limitations To prepare ISO and SAE passages for analysis, all titles, figures, tables, equations, headings, and references were removed from the 12 documents, since those elements were not intended to be processed by readability equations. This is a weakness of readability equations because the elements removed constitute a large fraction of the page space and are important to understanding each document.

Another limitation is that this study tested the readability of textual materials of two types of technical documents, not their application. We do not know whether people who have to follow the SAE or ISO standards can find what they need, understand what they find, and appropriately use what they find.

Suggestions for Further Research We would suggest that there be a further investigation of punctuation and word-level factors on predictions of readability, in particular, examining the punctuation and text elements examined in this paper. In addition, further study of how headings, figures, tables, and equations affect the overall grade reading level is desired. They constitute a large fraction of the page space in many documents, but their effect on readability is ignored in common readability equations. Finally, it would be useful to know how the readability of the set of standards examined fits into the bigger picture, for example, by examining the readability of highly cited standards, in either automotive or human factors engineering.

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